

# LEONARDO SANTOS

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[Homepage](#) ◊ [Google Scholar](#) ◊ [Github](#) ◊ [LinkedIn](#)

## EDUCATION

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### KTH Royal Institute of Technology

August 2025-Present, Sweden

Ph.D Student in the Division of Robotics Perception and Learning (RPL)

Supervisor: Prof. Iolanda Leite, Co-supervisor: Prof. Jana Tumova.

### UFMG Federal University of Minas Gerais

July 2025, Brazil

B.Sc. in Electrical Engineering, Emphasis in Robotics

GPA: 86/100

## SKILLS

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**Programming** Python, C++, MATLAB

**Technologies** ROS, RViz, Git,  $\LaTeX$

**Languages** English (fluent) and Portuguese (native)

## RESEARCH EXPERIENCE

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### Graduate Research Assistant

August 2025 – Present

KTH | [Prof. Iolanda Leite](#) | [RPL](#)

Stockholm, Sweden

- Investigating the use of Foundation Models and Reinforcement Learning from Human/AI Feedback for human-robot alignment.

### Robotics Institute Summer Scholar (RISS)

June 2024 – August 2024

Carnegie Mellon University | [Prof. Andrea Bajcsy](#) | [Intent Lab](#)

Pittsburgh, USA

- Developed a framework to update safety specifications online from human language feedback using a Vision Language Model.
- Applied the safety specifications in a downstream reachability controller for safe navigation and demonstrated the system in both hardware and simulation experiments. Co-first author [paper](#) under review to ICRA 2025 [1].
- More information and demos on the [project website](#).
- Technical skills:** Python, ROS.

### Research Assistant

September 2023 – January 2024

KTH | [Prof. Jana Tumova](#) | [RPL](#)

Stockholm, Sweden

- Studied risk-aware trajectory optimization for safe navigation among humans.
- Implemented an MPPI-based trajectory optimizer to minimize a CVaR risk measure in belief space for trajectory planning under uncertainty.
- Technical skills:** Python, PyTorch.

### Robotics Institute Summer Scholar (RISS)

June 2023 – August 2023

Carnegie Mellon University | [Prof. Sebastian Scherer](#) | [AirLab](#)

Pittsburgh, USA

- Developed a framework based on Random Finite Sets that mathematically unifies search and track objectives for informative path planning.
- Contributed to both theoretical and implementation developments.
- Co-first author [paper](#) published at 2025 AIAA SciTech Forum [2].
- Technical skills:** C++, MATLAB, ROS.

## Research Assistant

UFMG | [Prof. Douglas Macharet](#) | [VeRLab](#)

February 2021 – May 2023

Belo Horizonte, Brazil

- Developed a planning pipeline for communication-constrained multi-robot systems. The pipeline is based on virtual structures and guarantees that communication between robots is preserved by construction. Demonstrated in simulation and in a simple hardware experiment. First author [paper](#) under review at IEEE Latin America Transactions [3].
- Ran and evaluated experiments in a project that uses a heterogeneous multi-robot system for warehouse management. Second author [paper](#) published at LARS 2022 [4].
- **Technical skills:** Python, ROS.

## PUBLICATIONS

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- [1] L. Santos\*, Z. Li\*, L. Peters, S. Bansal<sup>†</sup>, and A. Bajcsy<sup>†</sup>, “Updating robot safety representations online from natural language feedback,” *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [2] L. Santos\*, B. Moon\*, S. Scherer, and H. Van Nguyen, “Unisat: Unified-objective belief model and planner to search for and track multiple objects,” in *AIAA SCITECH 2025 Forum*, 2025, p. 2114.
- [3] L. Santos, C. C. G. Ribeiro, and D. G. Macharet, “Communication backbone reconfiguration with connectivity maintenance,” *arXiv preprint arXiv:2409.16851*, 2024.
- [4] C. C. G. Ribeiro, L. H. M. C. dos Santos, and D. G. Macharet, “Collaborative ugv/uav path planning for inventory management in warehouses,” in *2022 Latin American Robotics Symposium (LARS)*, 2022, pp. 121–126.

\* Indicates equal contribution, † indicates equal advising.

## AWARDS, GRANTS AND SCHOLARSHIPS

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|--------------------------------------------------------------------------------------------------|------------------------|
| <b>IEEE RAS Travel Grant to attend ICRA 2025</b>                                                 | <i>Spring 2025</i>     |
| <b>Google Undergraduate Research Scholarship</b> , 1 of 3 scholars at CMU                        | <i>Summer 2024</i>     |
| <b>Minas Mundi Study Abroad</b> , selected UFMG student                                          | <i>Fall 2023</i>       |
| <b>Mobility21 Undergraduate Research Scholarship</b> , 1 of 3 scholars at CMU                    | <i>Summer 2023</i>     |
| <b>National Finalist</b> , top 3 Overall, <a href="#">2nd SARC-BARINet Aerospace Competition</a> | <i>Fall 2022</i>       |
| <b>Top 15 Papers Award</b> out of 80 submissions, LARS 2022                                      | <i>Fall 2022</i>       |
| <b>Minas Gerais Research Support Foundation Scholarship</b>                                      | Fall 2022, Spring 2023 |
| <b>Brazilian National Council for R&amp;D Scholarship</b>                                        | Fall 2021, Spring 2022 |

## SERVICE AND OUTREACH

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### Reviewer

July 2025 – Present

- NeurIPS 2025 (Position paper track); ACM/IEEE HRI 2026.

### Member of Admissions Committee

January 2025

[Carnegie Mellon University's Robotics Institute Summer Scholars](#)

- Reviewed applications for the Robotics Institute Summer Scholars (RISS) program at Carnegie Mellon University (CMU).
- RISS is a funded summer research program that brings national and international students to do research at the CMU Robotics Institute.

### Co-Lead Organizer

June 2023 – August 2024

[RoboLaunch](#)

Pittsburgh, USA

- Helped organize a robotics seminar series aimed at introducing undergraduate students to robotics through a series of accessible talks from speakers from academia and industry.
- Led logistics involving speaker invitations, advertisement, and general team coordination.

- During the time I was involved, we hosted speakers from CMU, MIT, the University of Washington, the University of Adelaide, ETH Zurich, among others.

**Co-Founder and Webmaster**  
*UFMG IEEE Student Branch*

September 2024 – July 2025  
Belo Horizonte, Brazil

- Organized events to promote and advertise research at UFMG, including seminars on microelectronics, AI, robotics, and biomedical engineering; research partnerships between Brazilian and French universities.
- Was responsible for everything related to branding, external communications, and social media presence.

SOCIETY MEMBERSHIPS

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|--------------------------------|-----------------------|
| <b>IEEE RAS Student Member</b> | <i>2025 – Present</i> |
| <b>IEEE Student Member</b>     | <i>2024 – Present</i> |